

vengeanceaiUSC-LBOMODEL1 — Assumptions List (Commentary + Tab/Cell Map)

AI-SDR LBO assumptions with workbook locations. Primary math is LBO/operating (not DCF). **Excel:**

https://raw.githubusercontent.com/vengeanceaiUSC/Rainmaker/cursor/vengeanceaiusclbomodel1-44dc/vengeanceaiUSC_LBOMODEL1.xlsx

#	Assumption / Value	Engine + Input	Math location	Commentary
1	Revenue Growth 10% per year (unchanged by AI)	LBO (not DCF) Input: Assumptions_Drivers!C5	AI_Operating!D5:H5 Revenue_t = Revenue_{t-1} × (1 + Assumptions_Drivers!C5). Same growth in AI and Base cases.	Annual growth remains stable at ten percent. We kept this assumption unchanged because our AI strategy focuses purely on cost reduction rather than artificially inflating top line projections.
2	Gross Margin 80%	LBO (not DCF) Input: Assumptions_Drivers!C6	AI_Operating!D6:H6 (COGS) COGS_t = Revenue_t × (1 – Assumptions_Drivers!C6); Gross Profit_t = Revenue_t × C6.	Expansion reached precisely eighty percent. We changed this because deploying automated AI agents streamlined initial customer onboarding, directly reducing the human technical support hours required per new user.
3	Sales Payroll –15% in Year 1	LBO (not DCF) Input: Assumptions_Drivers!C7	AI_Operating!D7 (Sales payroll Y1) Payroll_Y1 = Assumptions_Drivers!C22 × (1 – C7). Baseline payroll = C22.	Baseline costs dropped by fifteen percent immediately. We changed this to reflect the aggressive termination of outbound human development representatives, instantly eliminating their recurring annual base salary expenses.
4	Variable Commissions –20% of baseline commission pool from Y1	LBO (not DCF) Input: Assumptions_Drivers!C8	AI_Operating!D8:H8 (Commissions) Commissions_AI = Assumptions_Drivers!C23 × (1 – C8); Base keeps full C23 pool scaled with revenue.	Rates decreased significantly starting in year one. Replacing human representatives with AI meant fewer outbound deals required traditional payouts, allowing the company to retain higher profit per sale.
5	AI Infrastructure \$15 million fixed / year	LBO (not DCF) Input: Assumptions_Drivers!C9	AI_Operating!D10:H10 (AI infrastructure) AI_cost_t = Assumptions_Drivers!C9 each forecast year (AI case only). EBITDA reduces by this amount.	A new fixed fifteen million dollar expense was added. We included this vital change to cover enterprise software licenses, API usage limits, and compute power running the AI.
6	Sales Growth (S&M; expense growth) 2% per year from Year 2	LBO (not DCF) Input: Assumptions_Drivers!C10	AI_Operating!E7:H7 / E8:H8 (payroll & commissions Y2–Y5) For t≥2: Payroll_t = Payroll_{t-1} × (1 + Assumptions_Drivers!C10); same for commissions.	Expense growth was permanently capped at two percent. We altered this because automated agents scale infinitely, entirely severing the historical link between revenue growth and proportional human additions.
7	Administrative Costs (G&A;) Locked at baseline % of revenue	LBO (not DCF) Input: Assumptions_Drivers!C11	AI_Operating!D12:H12 (G&A;) G&A_t = Revenue_t × Assumptions_Drivers!C11. Identical % in AI and Base cases.	General overhead remained locked at baseline percentages. We avoided changing this because the value creation strategy strictly impacts the active sales floor without altering corporate finance or compensation.
8	Capital Expenditures –5% vs baseline CapEx rate	LBO (not DCF) Input: Assumptions_Drivers!C12	AI_Operating!D19:H19 (CapEx) CapEx_AI = Revenue_t × Assumptions_Drivers!C24 × (1 – C12); Base uses C24 only.	Hardware spending was reduced by five percent. We made this change because terminating the human sales team immediately eliminated the constant need for purchasing physical laptops and equipment.
9	Working Capital Receivables / WC requirement –10% vs base	LBO (not DCF) Input: Assumptions_Drivers!C13	AI_Operating!D20:H20 (ΔNWC) ΔNWC_AI = ΔRevenue_t × Assumptions_Drivers!C25 × (1 – C13).	Receivables collection periods improved slightly overall. Implementing automated AI billing reminders accelerated cash collections from late customers, lowering total working capital requirements and freeing extra cash for debt.

10	Term Loan A interest SOFR + 4.00%	LBO (not DCF) Input: Assumptions_Drivers!C14 / C15	Debt_Sweep_AI!C12 (TLA rate); Interest on AI_Operating!D17:H17 $TLA_rate = Assumptions_Drivers!C14 + C15 (=C27)$. $Interest_TLA = TLA_beg \times TLA_rate$.	The interest rate remained static at SOFR plus four. This was unchanged because the macroeconomic lending environment and the base company credit profile strictly dictate senior pricing.
11	Term Loan B interest SOFR + 5.00%	LBO (not DCF) Input: Assumptions_Drivers!C14 / C16	Debt_Sweep_AI!D12 (TLB rate); Interest on AI_Operating!D17:H17 $TLB_rate = Assumptions_Drivers!C14 + C16 (=C28)$. $Interest_TLB = TLB_beg \times TLB_rate$.	Pricing was maintained exactly at SOFR plus five. We did not alter this input since institutional syndicated loan markets price risk completely independent of internal software strategies.
12	Mandatory Amortization 1% of initial principal / year	LBO (not DCF) Input: Assumptions_Drivers!C17	Debt_Sweep_AI!D5:D9 (Mandatory 1%) $Mandatory_t = Initial_Debt \times Assumptions_Drivers!C17$ (floor paydown before optional sweep).	Principal repayment schedules were held constant at one percent. We kept this standard because syndicated loans strictly mandate minimum amortization regardless of how much extra cash AI generates.
13	Cash Sweep 100% of excess cash to debt	LBO (not DCF) Input: Assumptions_Drivers!C18	Debt_Sweep_AI!E5:E9 (Total paydown) $Optional_sweep_t = C18 \times \max(0, FCF_t - Mandatory_t)$; $Debt_end = Debt_beg - Mandatory - Optional_sweep$.	One hundred percent of excess cash targets debt paydown. This remained unchanged to ensure all newly generated margins from AI savings aggressively reduce leverage and maximize sponsor returns.
14	Tax Rate 21%	LBO (not DCF) Input: Assumptions_Drivers!C19	AI_Operating!D18:H18 (Net income) $Tax_t = \max(0, EBIT_t - Interest_t) \times Assumptions_Drivers!C19$; $NI_t = EBT_t - Tax_t$.	The corporate tax rate remained steady at twenty one percent. We did not change this assumption because federal statutory tax obligations remain unaffected by internal operating margin improvements.
15	Exit Multiple 13.0x EV / EBITDA	LBO (Ex LBO!D17; not DCF) Input: Assumptions_Drivers!C20	LBO!D17 (also Strategy Summary!D10) $Exit_EV = Y5_EBITDA \times Assumptions_Drivers!C20$; $Exit_Equity = Exit_EV - Debt_end$; MOIC / IRR from that.	The valuation multiple was securely locked at thirteen. We kept this exact multiple consistent across scenarios to prove equity value creation stemmed purely from AI operational improvements alone.

DCF note: These items drive the LBO operating engine (AI_Operating / Debt_Sweep_AI / LBO exit). They are not the primary inputs on the DCF tab.

PDF raw link: https://raw.githubusercontent.com/vengeanceai/USC/Rainmaker/cursor/vengeanceaiusclbomodel1-44dc/LBO/output/LBOMODEL1_ASSUMPTIONS_LIST.pdf

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